



A full-stack platform for managing client projects, automated quotations & seamless project delivery.

Project Title	ProjectPort – Project Management Platform
Team Name	IBM PBEL (Web Dev)
Team Members	Aryan Gupta
Course	Full Stack Web Development
Guide / Mentor	Prof. Naina Ma'am
Institution	IBM PBEL
Date	February 2026

Introduction

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Background

ProjectPort is an intelligent, full-stack project management and client interaction platform. Designed specifically for development agencies, freelancers, and project-based businesses, it streamlines the entire project lifecycle — from the first client inquiry all the way to final delivery.

Built with React 18 on the frontend and Node.js/Express on the backend, with a MySQL database, ProjectPort automates quotations, manages leads, and enables real-time collaboration between clients, admins, and developers.

Why This Topic?

Rising demand for remote & freelance project work

No all-in-one platform for inquiry → delivery workflow

Manual quotation processes waste time & cause errors

Clients lack visibility into project progress & payments

Problem Domain

Development agencies lose leads due to slow quote turnarounds and fragmented communication channels.

Problem Statement

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Development agencies and freelancers lack a unified digital platform that handles the full project lifecycle — from capturing client inquiries and generating accurate quotations to managing milestones, payments, and delivery tracking — leading to revenue loss and poor client experience.

❓ Fragmented Workflow

Client inquiries, pricing, project tracking, and payments managed across disconnected tools — email, spreadsheets, WhatsApp.

❓ Manual Quotation Errors

Manually calculating project prices is slow, inconsistent, and error-prone, leading to underpricing or client disputes.

❓ Zero Visibility

Clients have no real-time view of project status, milestone progress, or payment schedules after signing up.

❓ Solving this saves hours per lead, prevents revenue loss, and delivers a professional client experience.

Objectives

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Main Goal

Build an intelligent, automated full-stack project management platform that unifies client inquiry, quotation, delivery and payment tracking in one seamless product.

01 Dynamic Project Wizard

Multi-step intelligent form with real-time price calculation for web, mobile, e-commerce & custom projects.

02 Automated Quotation Engine

Excel-driven pricing rules → JSON config for instant, accurate cost estimation with detailed breakdowns.

03 Complete Lead Management

Capture, review & convert leads to active projects with one-click automated client account creation.

04 Comprehensive Dashboards

Client & Admin dashboards with real-time project status, payment tracking and business analytics.

05 Milestone & Payment System

Milestone tracking, payment schedules, invoice management, document sharing and activity logs.

06 JWT Security & RBAC

Role-based access control for Clients, Admins & Developers with bcrypt password hashing.

Literature Review / Existing Systems

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Platform	What It Offers	Key Limitation
Upwork	Large talent marketplace	20% fee; poor fit for agency-client workflows; no project wizard
Fiverr	Gig-based service marketplace	Gig model not suited for complex custom projects; no inquiry tracking
Freelancer.com	Competitive bidding platform	Overcrowded; spam bids; no automated pricing; no milestone control
Jira / ClickUp	Internal project management	No client-facing portal; no lead capture; no quotation engine
Custom Emails	Used by most small agencies	Zero automation; manual pricing; no tracking; unprofessional experience

Research References:

Statista (2024): Freelance Platform Market Report • IEEE (2022): Microservice Architecture Patterns • Sommerville, I. (2016): Software Engineering, 10th Ed.

Proposed System – ProjectPort

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ProjectPort replaces fragmented tools with a single unified platform covering the complete journey: Inquiry → Quotation → Lead Management → Project Execution → Milestone Tracking → Payment → Delivery.

CLIENT

- ▶ Submit project inquiries via wizard
- ▶ Receive instant automated quotations
- ▶ Track project progress in real-time
- ▶ Manage payments & view invoices

ADMIN

- ▶ View & qualify all incoming leads
- ▶ One-click convert lead → project
- ▶ Assign teams, set milestones
- ▶ Monitor analytics & pipeline

DEVELOPER

- ▶ Access assigned project tasks
- ▶ Update milestone progress
- ▶ Team collaboration tools
- ▶ (Coming Soon — full feature set)

KEY IMPROVEMENT: Single platform replaces 5+ disconnected tools — saving time, reducing errors and boosting client trust.

System Architecture / Design

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🔗 CLIENT LAYER — React 18 + Vite | Tailwind CSS | React Router v6 | Context API



🔗 API GATEWAY — Express.js 4.x | REST API | CORS | JWT Auth Middleware | Helmet Security



Auth Service
JWT + bcrypt

Lead Service
Inquiry capture

Project Service
Milestones

Notification
Service



🔗 DATA LAYER — MySQL 8.0 | mysql2 driver | Parameterized queries | Connection Pooling

Data Flow:

Client Browser → React Router → Page Component → Fetch API → Express Route → Auth Middleware → MySQL → JSON Response → State Update → UI Render

MODULES: User Management | Lead Pipeline | Project Management | Milestone Tracker | Payment Engine | Notifications | SEO/PWA

Technologies Used

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Frontend

React 18.2

UI Framework

Vite

Build Tool

Tailwind CSS 3.x

Styling

React Router v6

Routing

React Helmet Async

SEO

Lucide React

Icons

Backend

Node.js 20.x

Runtime

Express.js 4.x

Framework

JWT

Auth

bcrypt.js

Password Hashing

dotenv

Config

CORS / Helmet

Security

Database & DevOps

MySQL 8.0

Primary DB

mysql2

DB Driver

Git & GitHub

Version Control

Vercel

Frontend Hosting

Render

Backend Hosting

npm

Package Manager

Implementation

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Planning & Setup

ERD design, DB schema (schema.sql), API blueprint, folder architecture, Git repo setup

02

Authentication Layer

JWT login/register endpoints, bcrypt hashing, role-based middleware (Client / Admin)

03

Lead & Quotation System

Project wizard (multi-step form), real-time SidebarPricing, Quotation Engine (JSON config)

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Project Management

Lead → Project conversion, milestone CRUD, payment schedules, Admin & Client dashboards

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Notifications & SEO

Real-time in-app notifications, React Helmet SEO, sitemap.xml generator, PWA manifest

📦 Key Features Built

Results & Output

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100%

Test Pass Rate

< 2s

Avg Page Load Time

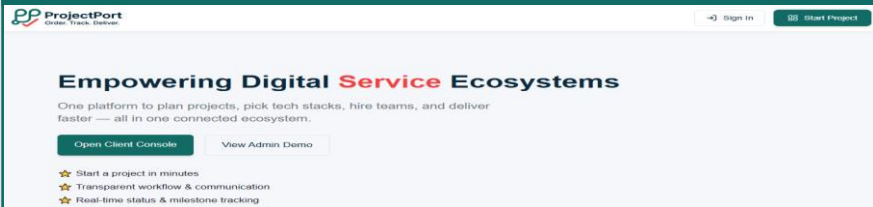
95+

Lighthouse Score

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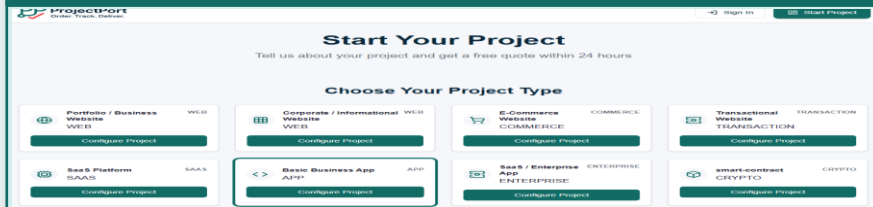
User Roles Supported

Landing / Home Page



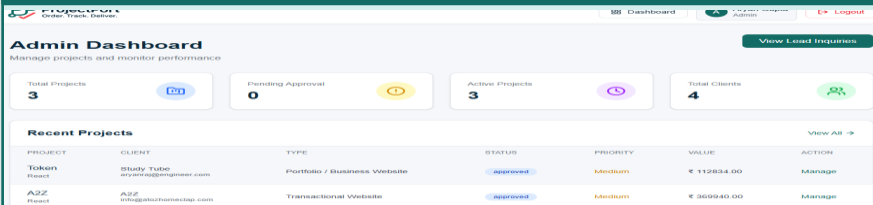
The landing page features the ProjectPort logo and tagline "Order Track. Deliver." with links for "Sign In" and "Start Project". The main heading is "Empowering Digital Service Ecosystems", followed by a subtext: "One platform to plan projects, pick tech stacks, hire teams, and deliver faster — all in one connected ecosystem." Below this are two buttons: "Open Client Console" and "View Admin Demo". A list of features includes: "Start a project in minutes", "Transparent workflow & communication", and "Real-time status & milestone tracking".

Client Project Wizard



The wizard starts with the heading "Start Your Project" and a subtext: "Tell us about your project and get a free quote within 24 hours." It then prompts the user to "Choose Your Project Type" and displays a grid of eight options, each with a "Configure Project" button. The options are: Portfolio / Business Website (WEB), Corporate / Informational Website (WEB), E-Commerce Website (COMMERCE), Transactional Website (TRANSACTION), SaaS Platform (SAAS), Basic Business App (APP), SaaS / Enterprise App (ENTERPRISE), and Smart-contract (CRYPTO).

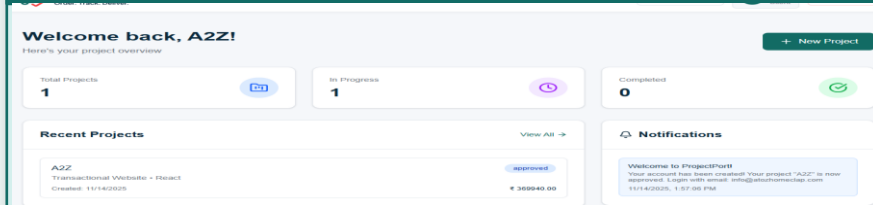
Admin Leads Dashboard



The dashboard includes a navigation bar with "Dashboard", "Admin", and "Logout" links, and a "View Lead Inquiries" button. It displays four summary cards: "Total Projects" (3), "Pending Approval" (0), "Active Projects" (3), and "Total Clients" (4). Below these is a "Recent Projects" table with columns for Project, Client, Type, Status, Priority, Value, and Action.

PROJECT	CLIENT	TYPE	STATUS	PRIORITY	VALUE	ACTION
Token React	Study Tube aryan@studytube.com	Portfolio / Business Website	Approved	Medium	₹ 112834.00	Manage
A2Z React	A2Z info@studymarkup.com	Transactional Website	Approved	Medium	₹ 366940.00	Manage

Client Dashboard



The dashboard greets the user with "Welcome back, A2Z!" and "Here's your project overview". It features a "+ New Project" button and three summary cards: "Total Projects" (1), "In Progress" (1), and "Completed" (0). Below these is a "Recent Projects" table and a "Notifications" section.

PROJECT	STATUS	VALUE
A2Z Transactional Website - React Created: 11/14/2025	Approved	₹ 366940.00

Notifications

Welcome to ProjectPort
Your account has been created! Your project "A2Z" is now approved. Login via email: info@studymarkup.com
11/14/2025, 1:57:36 PM

Challenges Faced

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Technical Challenges

Automated Quotation Engine

Converting complex Excel pricing rules into a reliable real-time JSON-driven pricing system.

Role-Based Access Control

Seamlessly handling 3 user roles (Client/Admin/Dev) across frontend routes and backend middleware.

MySQL Relational Schema

Designing a normalized database that cleanly represents leads → projects → milestones → payments.

JWT Security in Production

Managing token expiry, refresh flows and httpOnly cookie options securely across environments.

Solution Strategy: Agile sprints + feature branching + weekly syncs + shared API contract documentation resolved coordination issues.

Team & Project Challenges

Parallel Development

Coordinating frontend and backend work simultaneously without blocking each other on API contracts.

Git Merge Conflicts

Integration conflicts during feature branch merges required a clear branching and PR review strategy.

Consistent UI/UX

Ensuring design coherence across pages built by different team members using a shared component library.

Scope Management

Keeping the feature set realistic for the timeline while planning the developer module for future sprints.

Future Scope & Roadmap

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Email Notifications

Planned

SMTP integration for critical event alerts — project updates, payment reminders, milestone completions.

File Upload & Docs

Planned

Attach briefs, designs and deliverables directly to projects using cloud storage (AWS S3).

Payment Gateway

Planned

Razorpay / Stripe integration for in-platform payments, automated receipts and GST invoices.

Mobile App

Roadmap

React Native apps for iOS & Android enabling on-the-go project management and push notifications.

AI Quotation Engine

Roadmap

ML-based intelligent pricing using historical project data — smarter, more accurate estimates.

Advanced Analytics

Roadmap

Revenue dashboards, lead conversion funnels, team performance metrics and exportable PDF/Excel reports.

CONCLUSION

ProjectPort successfully demonstrates how a focused student team can design and implement an enterprise-grade, full-stack SaaS product using modern open-source technologies.

*The platform is live at
projectsport.vercel.app*

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📋 Key Achievements

- ✓ Built a complete Lead → Quotation → Project → Payment workflow
- ✓ Automated quotation engine with Excel-driven JSON pricing config
- ✓ Role-based access: Client, Admin & Developer portals
- ✓ JWT authentication with bcrypt + parameterized SQL queries
- ✓ SEO optimized with React Helmet, sitemap & PWA manifest
- ✓ Deployed live on Vercel with CI/CD via GitHub
- ✓ Responsive mobile-first design with Tailwind CSS

References

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📖 Books

- Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.
- Flanagan, D. (2020). JavaScript: The Definitive Guide (7th ed.). O'Reilly Media.
- Martin, R. C. (2017). Clean Architecture. Prentice Hall.

📖 Official Documentation & Websites

- React.js Documentation – <https://react.dev>
- Node.js Official Site – <https://nodejs.org> | Express.js – <https://expressjs.com>
- MySQL 8.0 Reference Manual – <https://dev.mysql.com/doc>
- Tailwind CSS Docs – <https://tailwindcss.com/docs> | Vite – <https://vitejs.dev>
- Vercel Deployment Docs – <https://vercel.com/docs>

📖 Research Papers

📖 ProjectPort — Built with React, Node.js, Express & MySQL | github.com/aryanony/ProjectPort